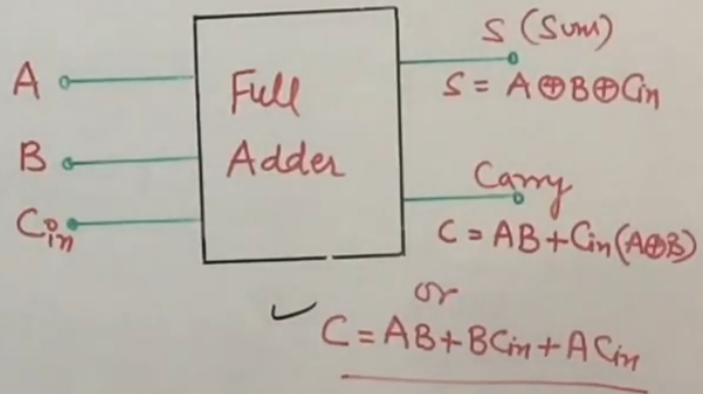
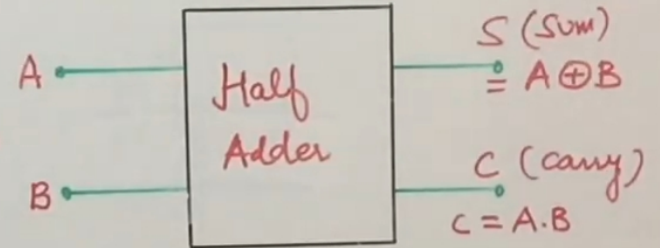
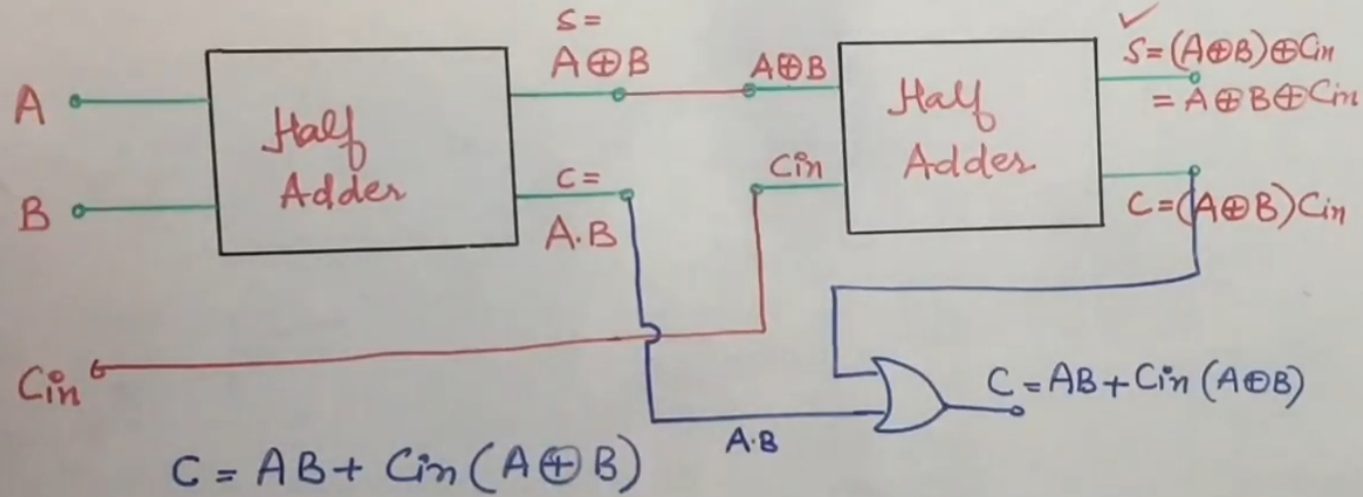
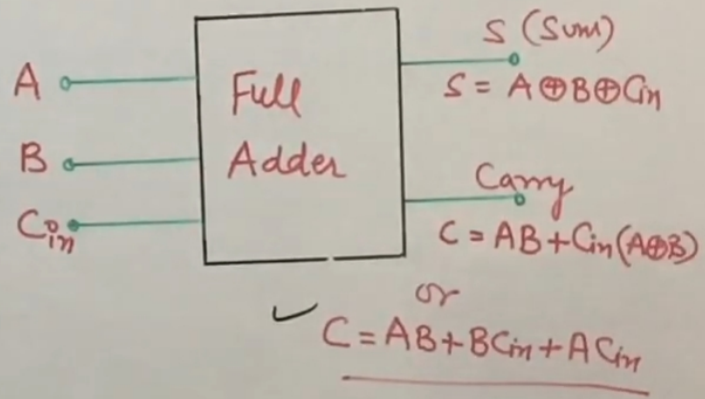
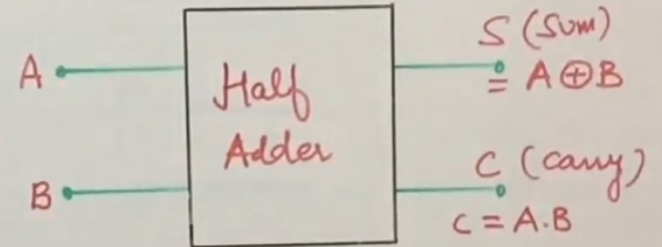
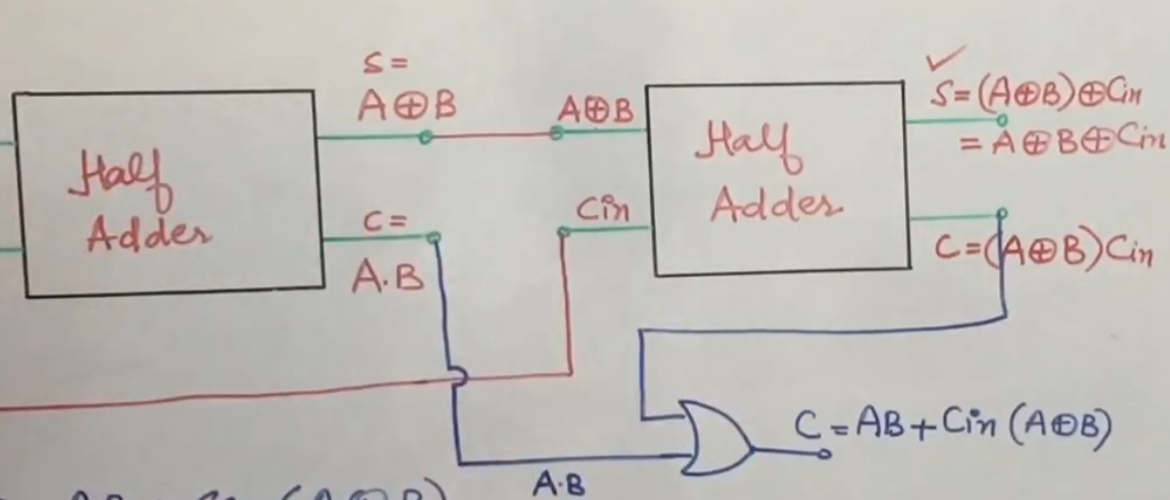


Full Adder Using Half Adder



Full Adder Using Half Adder



$$\begin{aligned}
 C &= AB + C_{in}(A \oplus B) \\
 &= AB + C_{in}(A\bar{B} + \bar{A}B) \\
 &= AB + A\bar{B}C_{in} + \bar{A}BC_{in} \quad [\because (A + \bar{A})B = A + B] \\
 &= A(B + \bar{B}C_{in}) + \bar{A}BC_{in} \\
 &= A(B + C_{in}) + \bar{A}BC_{in} \\
 &= AB + AC_{in} + \bar{A}BC_{in} \\
 &= AB + C_{in}(A + \bar{A}B) = AB + C_{in}(A + B) = AB + AC_{in} + BC_{in}
 \end{aligned}$$